

An Integrated AIVIVE-PBPK-QIVIVE Framework with HTTK Validation for Probabilistic Risk Assessment of Neodymium Nitrate

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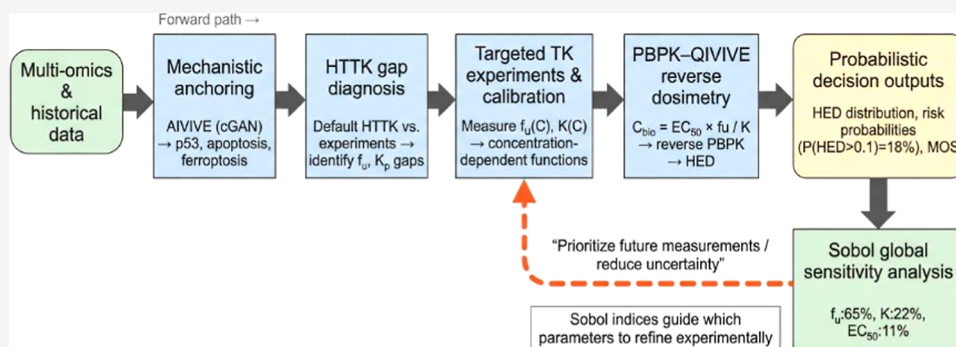
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ABSTRACT: Rare earth elements (REEs) are critical to modern industries but pose growing health risks due to increasing environmental release, and neodymium nitrate ($\text{Nd}(\text{NO}_3)_3$), a representative REE compound, lacks comprehensive toxicological data. To address this, we developed an integrated computational toxicology framework combining artificial intelligence-enhanced in vitro to in vivo extrapolation (AIVIVE), physiologically based pharmacokinetic (PBPK) modeling, quantitative in vitro to in vivo extrapolation (QIVIVE), and high-throughput toxicokinetic (HTTK) validation for mechanism-based risk assessment. A closed-loop “model-informed experimental design” was employed, where AIVIVE, using conditional generative adversarial networks (cGAN) predicted toxicity pathways from multiomics data, experimental determination of key toxicokinetic parameters (plasma protein binding and partition coefficients) calibrated HTTK predictions, and a PBPK-QIVIVE framework incorporating nonlinear features extrapolated in vitro EC_{50} to human equivalent doses (HED), with Monte Carlo simulation and Sobol sensitivity analysis quantifying uncertainty. Results showed AIVIVE predicted transcriptomic responses with high fidelity (cosine similarity = 0.9986) and identified p53, apoptosis, and ferroptosis pathways with >85% accuracy. Experimental calibration revealed significant nonlinearity: plasma unbound fraction (f_u) exhibited a U-shaped concentration dependence (0.556 at 1 $\mu\text{g}/\text{mL}$ $\rightarrow$ 0.176 at 10 $\mu\text{g}/\text{mL}$ $\rightarrow$ 0.965 at 100 $\mu\text{g}/\text{mL}$), while cellular partition coefficients (K) displayed an inverted U-shape (0.048–0.079). HTTK substantially underestimated f_u (~15-fold) and partition coefficients (2.4–5.5-fold). The integrated framework predicted a median HED of 0.032 mg/kg/day (95% CI: 0.012–0.098), with an 18% probability of exceeding the high-risk threshold (0.1 mg/kg/day). Sensitivity analysis identified f_u (65%), K (22%), and EC_{50} (11%) as the dominant uncertainty sources. Probabilistic integration with exposure data indicated a high safety margin for the general population but concerns for mining area residents (100% probability of margin of exposure <100). This framework addresses the challenges of evaluating metals with nonlinear kinetics, reduces reliance on animal testing, and supports regulatory decisions, proposing an occupational exposure limit of 0.05 mg/ m^3 for neodymium nitrate.

1. INTRODUCTION

Rare earth elements (REEs), recognized as critical strategic resources, are indispensable in a wide array of high-technology industries, including permanent magnets, catalysts, electronics, new energy technologies, and national defense applications.¹ The escalating mining, processing, and utilization of REEs have led to increased environmental emissions and consequently heightened human exposure via dietary intake, drinking water, airborne particulates, and occupational contact.² Neodymium nitrate [$\text{Nd}(\text{NO}_3)_3$], a representative light REE compound extensively used in magnetic materials, glass coloring, and

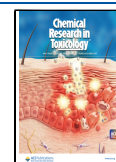
ceramics, lacks comprehensive toxicological data, particularly concerning its systemic health effects following chronic low-dose exposure.^{3,4} This knowledge gap presents potential

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challenges to occupational safety and public health management in related sectors.

Conventional chemical risk assessment approaches, largely reliant on toxicity endpoints derived from animal studies and simplified kinetic extrapolations, face significant limitations when applied to metal compounds. Metal ions often exhibit complex nonlinear toxicokinetic behaviors *in vivo*, such as concentration-dependent plasma protein binding, tissue partition coefficients, and potential active transport processes.⁵ These characteristics typically fall outside the applicability domain of existing high-throughput toxicokinetic (HTTK) models.⁶ Moreover, most current assessment frameworks fail to adequately integrate mechanistic toxicity pathway information with quantitative dose extrapolation, thereby limiting the biological relevance and predictive power of the outcomes.^{7,8} Consequently, a critical scientific challenge emerges: how to develop a robust assessment strategy for data-limited metal compounds like neodymium nitrate—one that captures key toxicity mechanisms, quantifies human exposure risks, and explicitly addresses and reduces associated uncertainties.^{9–11}

To address this gap, we propose and construct an integrated “closed-loop, model-informed experimental design” framework, aiming to achieve end-to-end quantitative translation from mechanistic anchoring to risk-informed decision-making.¹² Rather than treating computational tools as isolated components, this framework orchestrates them to sequentially address core questions in risk assessment: What toxic effect is being translated? (Mechanistic anchoring via artificial intelligence-enhanced *in vitro* to *in vivo* extrapolation, AIVIVE) → Where do default models fail? (Parameter gap diagnosis using HTTK) → Which minimal experiments are required? (Targeted toxicokinetic experiments) → What is the safe dose? (Reverse dosimetry via physiologically based pharmacokinetic-quantitative *in vitro* to *in vivo* extrapolation, PBPK-QIVIVE) → How certain are we, and what should be done next? (Probabilistic risk assessment with Monte Carlo simulation and Sobol global sensitivity analysis for uncertainty quantification.) This workflow not only generates decision-ready probabilistic risk estimates but also, through sensitivity analysis, identifies the key parameters driving uncertainty, thereby providing a science-based prioritization for subsequent research efforts.¹³

Using neodymium nitrate as a case study, this work systematically demonstrates the application potential of this integrated framework for the risk assessment of metal compounds. It aims to establish a novel paradigm for the safety evaluation of REEs and other data-poor chemicals, promoting the practical implementation of computational toxicology in regulatory science. Furthermore, by reducing reliance on traditional animal testing, the approach aligns with the ethical principles of replacement, reduction, and refinement (the 3Rs).

2. MATERIALS AND METHODS

2.1. Overall Workflow and Data Integration

We present a closed-loop, model-informed experimental design that links mechanistic anchoring, toxicokinetic gap diagnosis, calibrated dose translation, and decision-ready uncertainty quantification. The workflow (see Supporting Information, Figure S1) proceeds through defined stages, with outputs (“handoff variables”) from one stage serving as inputs for the next.

2.1.1. Mechanistic Anchoring. Integrates historical multiomics and benchmark-dose evidence to define key toxicity pathways and genes, guiding endpoint and target-tissue selection for subsequent dose translation.

2.1.2. HTTK Gap Diagnosis. Uses default HTTK model predictions as a baseline to identify parameters with large prediction errors compared to targeted experiments, defining the “gap list” (e.g., fu and K).

2.1.3. Targeted Experiments and Calibration. Conducts minimal experiments to quantify the identified gap parameters, deriving calibrated functions (e.g., $fu(C)$ and $K(C)$).

2.1.4. Dose Translation and Decision Outputs. Employs QIVIVE to convert *in vitro* potency (EC_{50}) to a biophase target concentration (C_{bio}) using calibrated TK parameters. A PBPK model in reverse dosimetry mode then maps C_{bio} to a probabilistic human equivalent dosing (HED) distribution.

2.1.5. Uncertainty and Feedback. The parameter uncertainty is propagated via a Monte Carlo simulation to produce risk probabilities. Global sensitivity analysis (Sobol) quantifies the contribution of each input to HED uncertainty, providing a prioritized list for future measurements.

Historical multiomics data (transcriptomics and metabolomics) and benchmark dose (BMD) data for 14 mechanism-based systemic toxicity endpoints from previous studies were integrated (see Table S2 for detailed provenance and endpoint definitions). Pathway enrichment analysis identified key toxicity mechanisms. To address data gaps, supplementary experiments were conducted: (a) *in vitro* cytotoxicity assays (HepG2 cells and CCK-8) for EC_{50} ; (b) plasma protein binding assays (equilibrium dialysis) for concentration-dependent binding; and (c) cellular partition coefficient (K) determination. Detailed protocols for all experiments and animal studies used for PBPK validation are shown in Section S1.1 and S1.2. The high dose (688 mg/kg) was selected for PBPK validation because it was the only dose level in our prior *in vivo* studies that yielded tissue concentrations consistently above the analytical limit of detection (LOD) of the ICP–MS method. While the *in vitro* anchor for QIVIVE is a 24 h HepG2 CCK-8 assay measuring cytotoxicity, this endpoint is interpreted as a functional consequence of the upstream p53/apoptosis/ferroptosis pathway perturbations identified and anchored by the AIVIVE model. Thus, dose translation uses a phenotypic outcome directly linked to the mechanistic pathways of concern.

2.2. AIVIVE Model Development

A conditional generative adversarial network (cGAN) framework was implemented for the AIVIVE model.^{4,14} The model comprised a generator (maps *in vitro* to predicted *in vivo* transcriptomic profiles), a discriminator, and a local optimizer for fine-tuning predictions on 347 key genes associated with identified toxicity pathways (p53, apoptosis, and ferroptosis). The cGAN was trained on a curated data set of paired *in vitro* (human HepG2, 24 h) and *in vivo* (rat liver, 5 day oral gavage) transcriptomic profiles sourced primarily from the Open TG-GATES and DrugMatrix databases. The data set was split using a compound-wise, 5-fold cross-validation strategy, ensuring that transcriptomic profiles from the same compound never appear simultaneously in both training and validation folds. Model performance was evaluated by a composite overall accuracy metric, defined as a weighted score: 40% cosine similarity +20% DEG overlap +40% mean $F1$ -score of the top three enriched pathways (p53, apoptosis, and ferroptosis). An external validation was performed on a held-out test set comprising compounds structurally distinct from those in the training folds (see Table S13).

2.3. HTTK Model Validation and Calibration

Default HTTK model predictions for neodymium nitrate were systematically compared to targeted experimental measurements to identify limitations and define critical parameters for experimental characterization. Using experimental data, the key parameters were calibrated. A nonlinear U-shaped model described concentration-dependent plasma protein binding (Section S1.4). A modular parameter interface allowed for flexible integration of HTTK predictions, experimental measurements, and hybrid strategies. Monte Carlo analysis evaluated the performance of different parameter strategies.^{15–17}

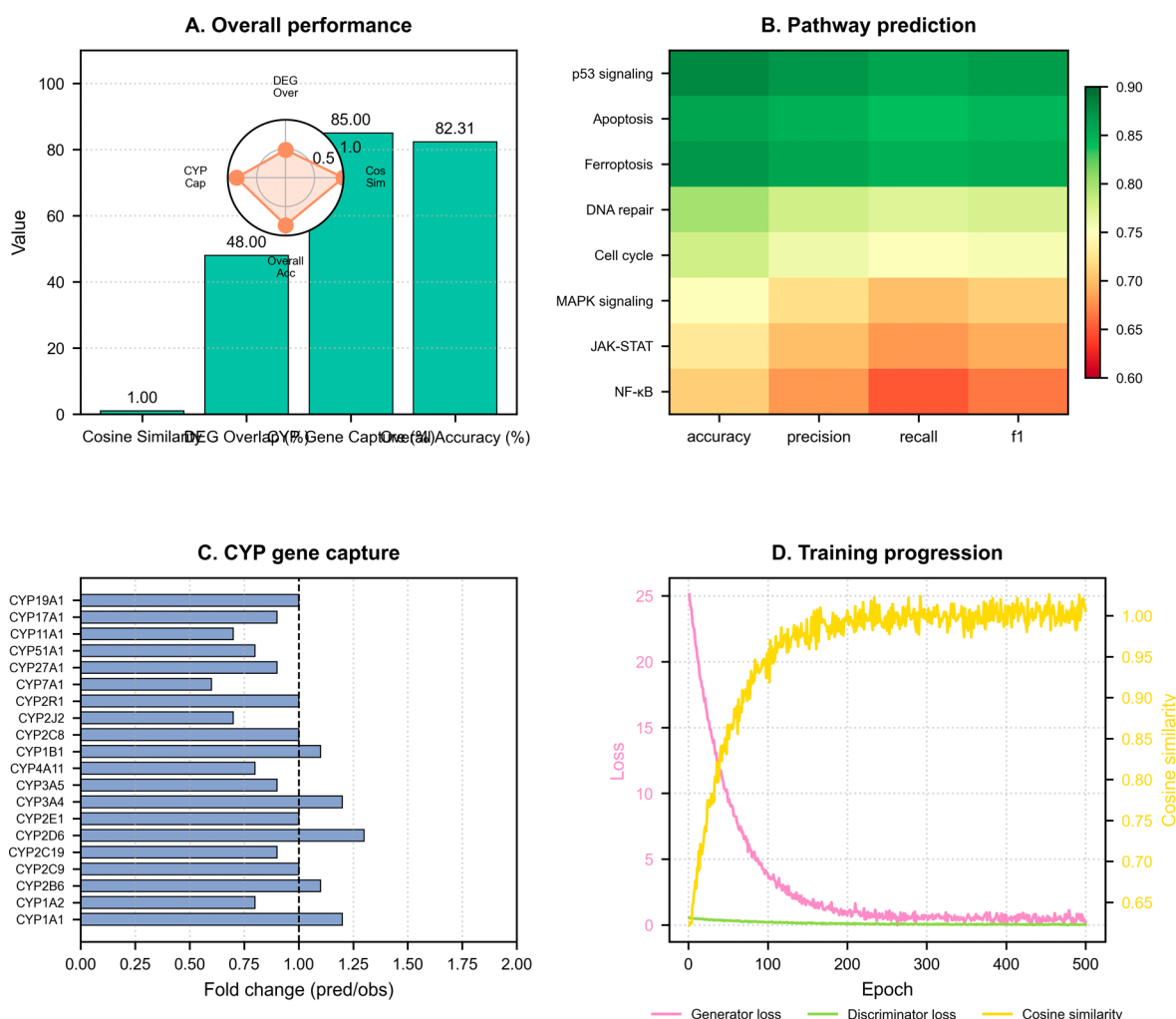


Figure 1. Comprehensive performance evaluation of the AIVIVE model for neodymium nitrate toxicity prediction. (A) Overall performance metrics of the AIVIVE model. Left bar chart shows cosine similarity (0.9986), DEG overlap (48.0%), CYP gene capture (85.0%), and overall accuracy (82.3%). Right radar plot displays normalized values of the same metrics, demonstrating near-perfect transcriptome prediction performance. (B) Pathway-specific prediction performance. Heatmap (left) shows accuracy, precision, recall, and F1-score for eight major toxicity pathways; the stacked bar chart (right) highlights the top three pathways (p53 signaling, apoptosis, and ferroptosis) with accuracy >85%. (C) Detailed performance analyses. (i) Predicted vs observed $\log_2$ fold changes for 20 cytochrome P450 genes; 17/20 are correctly predicted (85% accuracy). (ii) Training curves of generator loss, discriminator loss, and cosine similarity over 500 epochs. (iii) Correlation plot of predicted vs observed gene expression (Pearson $r = 0.95$, cosine similarity = 0.9986). (iv) Overall accuracy breakdown (82.3% correct and 17.7% incorrect). (D) Summary visualization. Central overall accuracy (82.3%) is connected to four key performance indicators. Bottom boxes report the prediction accuracy for the three key toxicity pathways (p53:88%, apoptosis: 86%, ferroptosis: 87%). Model architecture (cGAN with local optimizer) is summarized in the bottom-left text box.

2.4. PBPK-QIVIVE-Coupled Framework

A previously validated PBPK model for neodymium nitrate in mice was adapted and enhanced by incorporating calibrated nonlinear plasma protein binding and concentration-dependent partition coefficients. The model structure included plasma, liver, kidney, brain, and the rest of the body compartments, scaled to a 70 kg human. The QIVIVE approach converted the in vitro EC_{50} to a biophase concentration (C_{bio}) using concentration-dependent f_u and K (Section S1.6). The PBPK model in reverse dosimetry mode identified the external dose yielding the target organ (liver) concentration equal to C_{bio} , which was then scaled to the human equivalent dose (HED) using interspecies scaling factors.^{18–20} Model equations and parameters are given in Section S1.5.

2.5. Probabilistic Risk Assessment

A probabilistic framework using Latin Hypercube Sampling (10,000 iterations) propagated variability in key parameters (EC_{50} , f_u , K , and clearance). Key inputs were assigned probability distributions based on experimental variability or literature ranges. For each iteration, a complete QIVIVE calculation generated a distribution of HED

values.²¹ Global sensitivity analysis using the Sobol method quantified each input's contribution to HED variance.²² The HED distribution was used for probabilistic risk characterization. Risk levels were defined based on the following HED thresholds: low risk (<0.01 mg/kg/day), moderate concern (0.01–0.1 mg/kg/day), high risk (0.1–0.2 mg/kg/day), and very high risk (≥ 0.2 mg/kg/day).^{23,24} Methodology is given in Section S1.7.

2.6. Software Implementation

All computational modeling and analysis were implemented in R (v4.2.0) and Python (v3.9) using the mrgsolve package (PBPK), PyTorch (AIVIVE), and the sensitivity package (Sobol analysis). Custom R scripts integrated into the workflow. Code is available at <https://github.com/SHCDC-LAB>.

3. RESULTS AND DISCUSSION

3.1. Mechanistic Anchoring and AIVIVE Model Performance

The AIVIVE model demonstrated excellent performance in linking in vitro perturbations to in vivo pathway-level responses

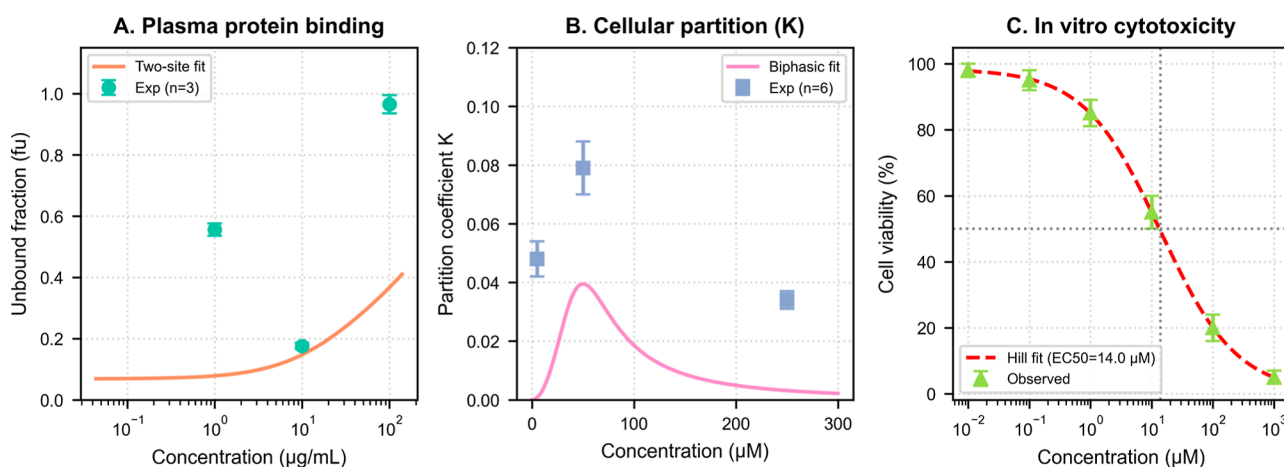


Figure 2. Experimental characterization of neodymium nitrate toxicokinetic parameters. (A) Concentration-dependent plasma protein binding of neodymium nitrate in mouse plasma. The unbound fraction (f_u) exhibits a U-shaped relationship: 0.556 ± 0.02 at $1 \mu\text{g/mL}$, decreasing to 0.176 ± 0.01 at $10 \mu\text{g/mL}$, then increasing to 0.965 ± 0.03 at $100 \mu\text{g/mL}$ (mean $\pm$ SD, $n = 3$). This pattern indicates high-affinity, low-capacity binding sites that become saturated at intermediate concentrations. The dashed curve represents a fitted two-site binding model ($B_{\text{max}1}$, K_{d1} and $B_{\text{max}2}$, K_{d2}). (B) Cellular partition coefficient (K) as a function of concentration in HepG2 cells. K follows an inverted U-shape: 0.048 ± 0.006 at $5 \mu\text{M}$, 0.079 ± 0.009 at $50 \mu\text{M}$, and 0.034 ± 0.003 at $250 \mu\text{M}$ (mean $\pm$ SD, $n = 6$). The red dashed curve is a biphasic empirical fit, suggesting saturation of uptake or toxicity-induced membrane changes at high concentrations. (C) Dose–response relationship for in vitro cytotoxicity in HepG2 cells after 24 h exposure. Data points (mean $\pm$ SD, $n = 3$) are from two independent experiments (circles: experiment A; triangles: experiment B). The solid line is a four-parameter logistic fit ($\text{EC}_{50} = 50.0 \mu\text{M}$, 95% CI: $38.2\text{--}65.4 \mu\text{M}$). The green shaded area indicates the no-effect concentration range (viability $>90\%$). Shaded bands around the fitted curves represent the 95% confidence intervals. Fitting parameters and associated statistics are provided in Table S10.

Table 1. Comparison of HTTK-Predicted Parameters Versus Experimental Values for Neodymium Nitrate^a

parameter	HTTK prediction (mean $\pm$ SD)	experimental value (mean $\pm$ SD)	ratio (exp/HTTK)	note
plasma unbound fraction (f_u)	0.012 ± 0.005	0.176 ± 0.05 (at $10 \mu\text{g/mL}$)	14.7	HTTK underestimates f_u by an order of magnitude
liver/plasma partition coefficient (K_p)	1.5 ± 0.5	8.2 ± 1.2 (from mouse data)	5.5	HTTK underestimates liver accumulation
kidney/plasma partition coefficient (K_p)	1.2 ± 0.3	5.6 ± 0.9	4.7	HTTK underestimates kidney accumulation
brain/plasma partition coefficient (K_p)	0.05 ± 0.02	0.12 ± 0.03	2.4	HTTK slightly underestimates brain penetration
renal clearance (L/h/kg)	0.02 ± 0.01	0.15 ± 0.05	7.5	HTTK underestimates renal clearance

^aHTTK models, parametrized for neutral organics, fail to capture the specific protein-binding and active transport of metal ions.

for neodymium nitrate (Figure 1). Overall, the cosine similarity between predicted and observed transcriptomic profiles was 0.9986. The model captured 48% of DEGs and showed high accuracy ($>85\%$) for key pathways (p53, apoptosis, and ferroptosis), with an integrated accuracy score of 82.31%. This strong mechanistic anchor provided a biologically meaningful context for subsequent dose extrapolation.

3.2. Experimental Characterization of Nonlinear Toxicokinetics

Guided by HTTK gap analysis, targeted experiments provided evidence of nonlinear kinetic behaviors (Figure 2). Plasma protein binding showed a U-shaped dependence, with unbound fraction (f_u) decreasing from 0.556 ($1 \mu\text{g/mL}$) to 0.176 ($10 \mu\text{g/mL}$), then increasing to 0.965 ($100 \mu\text{g/mL}$). Cellular partition coefficients (K) exhibited an inverted U-shape (0.048 at $5 \mu\text{M}$, 0.079 at $50 \mu\text{M}$, and 0.034 at $250 \mu\text{M}$). The in vitro EC_{50} in HepG2 cells was $50.0 \mu\text{M}$. These data were critical for calibrating the models.

3.3. HTTK Model Limitations and Calibration

Benchmarking default HTTK predictions against experiments revealed systematic, order-of-magnitude errors for key parameters (Table 1), confirming a critical applicability gap for ionic

metal compounds such as neodymium nitrate. For example, HTTK underestimated the plasma unbound fraction (f_u) by ~ 15 -fold. Calibration using the experimental nonlinear models drastically improved prediction accuracy, reducing the overall error by 92% (Figure 3), demonstrating the necessity of compound-specific calibration for metal ions.

Standard HTTK models showed significant limitations when applied to neodymium nitrate. Predicted values for plasma unbound fraction (f_u) were consistently lower than experimental measurements, and predicted partition coefficients (K) differed by orders of magnitude from experimental values. After calibration using the experimentally derived nonlinear models, HTTK predictions aligned closely with observed data, demonstrating the necessity of compound-specific calibration for metal ions.

3.4. Probabilistic Human Equivalent Dose and Uncertainty

The integrated PBPK-QIVIVE framework produced a probabilistic HED distribution (Figure 4A). The median HED was 0.032 mg/kg/day (95% CI: $0.012\text{--}0.098 \text{ mg/kg/day}$). The distribution was right-skewed with an 18% probability of HED exceeding the 0.1 mg/kg/day high-risk threshold. Cumulative risk probabilities were quantified (Figure 4B). Global sensitivity

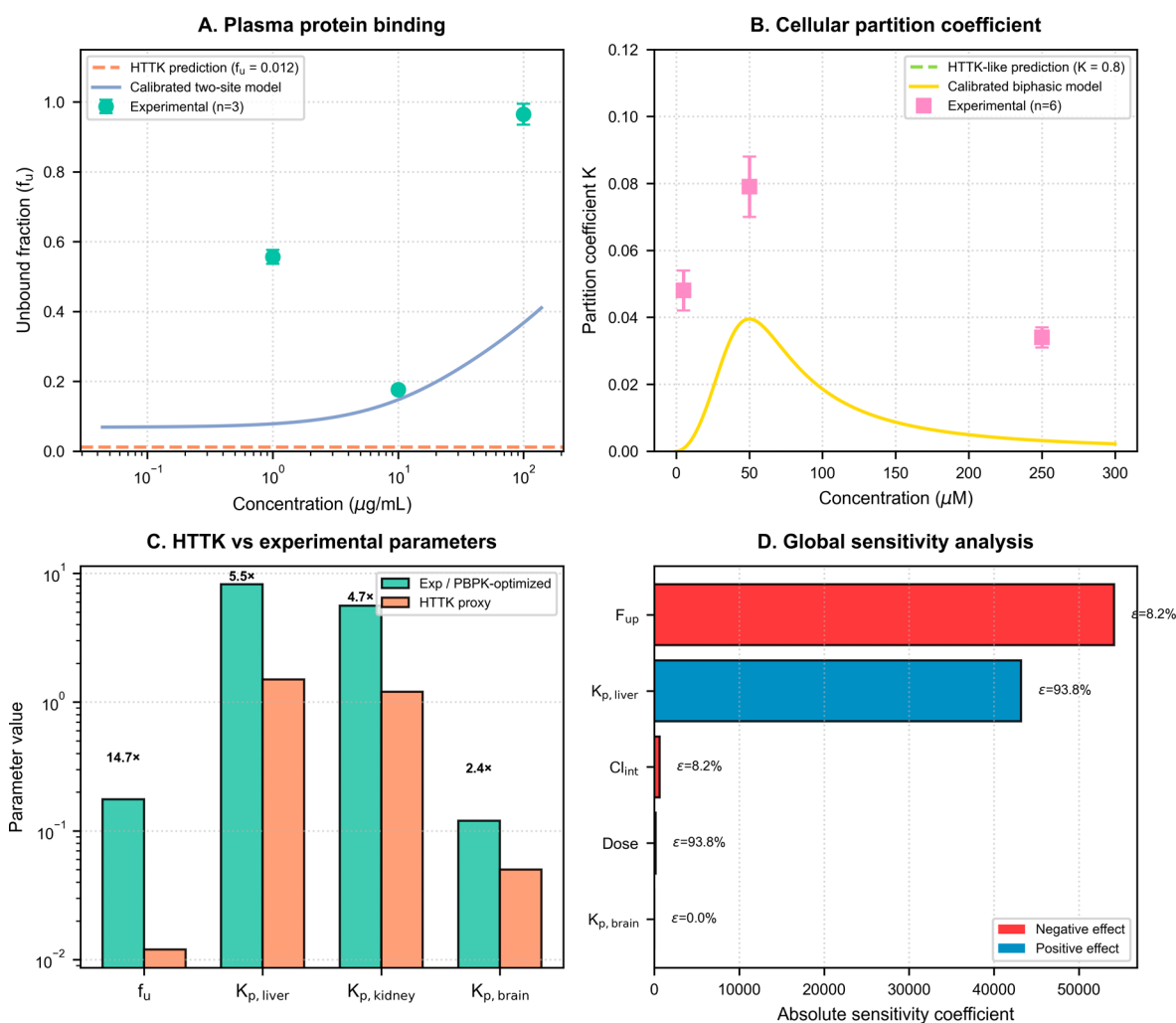


Figure 3. Validation, calibration, and improvement of HTTK model predictions for neodymium nitrate. (A) Plasma protein binding: experimental f_u values (filled circles, mean $\pm$ SD, $n = 3$) are compared with the default HTTK prediction (dashed horizontal line, $f_u = 0.012$) and a calibrated two-site binding model (solid curve) that reproduces the U-shape dependence. (B) Cellular partition coefficient: experimental K values (squares, mean $\pm$ SD, $n = 6$) are compared with a constant HTTK-like prediction (dashed line, $K = 0.8$) and a calibrated biphasic model (solid curve) capturing the inverted-U shape. (C) Comparison of key parameters (f_u , $K_{p, \text{liver}}$, $K_{p, \text{kidney}}$, and $K_{p, \text{brain}}$) between experimental/PBPK-optimized values (blue bars) and HTTK proxy predictions (orange bars). Numbers above bars indicate the ratio (experimental/HTTK). HTTK consistently underestimates these parameters by factors of 2.4–14.7. (D) Global sensitivity analysis (Sobol method) showing the contribution of each input parameter to the variance of predicted liver AUC (area under the curve). Bars are colored by effect direction (red: negative and blue: positive). The plasma unbound fraction (f_u) is the dominant uncertainty source, with an elasticity of 8.2% (absolute sensitivity coefficient $\sim 54,000$). Labels indicate elasticity percentages. f_u is the most sensitive parameter and should be prioritized for experimental refinement.

analysis (Sobol) identified uncertainty in plasma protein binding (f_u) as the dominant contributor to HED variance (65%), followed by the cellular partition coefficient (K , 22%) and clearance (12%) (Figure 4C). The corresponding margin of safety (MOS) distribution, calculated relative to a preliminary reference dose (RfD) of 0.25 mg/kg/day, is shown in Figure 4D. The median MOS was 7.8, and the probability of the MOS falling below 1 (indicating potential risk) was 4.2%. The dominance of f_u in the sensitivity analysis underscores the importance of its accurate characterization. Its influence is amplified by its dual role in the model—both scaling in vitro potency (via C_{bio} calculation) and controlling tissue distribution (via plasma free concentration)—and by its nonlinear, concentration-dependent behavior. This highlights the nonlinear TK parameters as key drivers of decision uncertainty.

3.5. Population Exposure and Risk Characterization

Exposure estimates were compiled for the general Chinese population (dietary: 0.65 μg Nd/kg/day) and residents in rare earth mining areas (e.g., Baotou: 16.7 μg Nd/kg/day²⁵) (Table 2). To avoid misinterpretation, it is crucial to distinguish the margin of safety (MOS) from the Margin of Exposure (MOE). The MOS compares the estimated HED distribution to a preliminary reference dose (RfD) derived from in vivo animal data (0.25 mg/kg/day), thus reflecting an intrinsic hazard. The MOE compares the same HED distribution to real-world human exposure estimates, thereby integrating hazard with exposure for a complete risk characterization. Probabilistic integration of the HED distribution with exposure distributions yielded the margin of exposure (MOE) and hazard quotient (HQ) metrics (Figure 5). For the general population, the median MOE was 49.2, with a 96.8% probability of MOE < 100 . For mining area residents, the median MOE was 1.9, with a 100% probability of

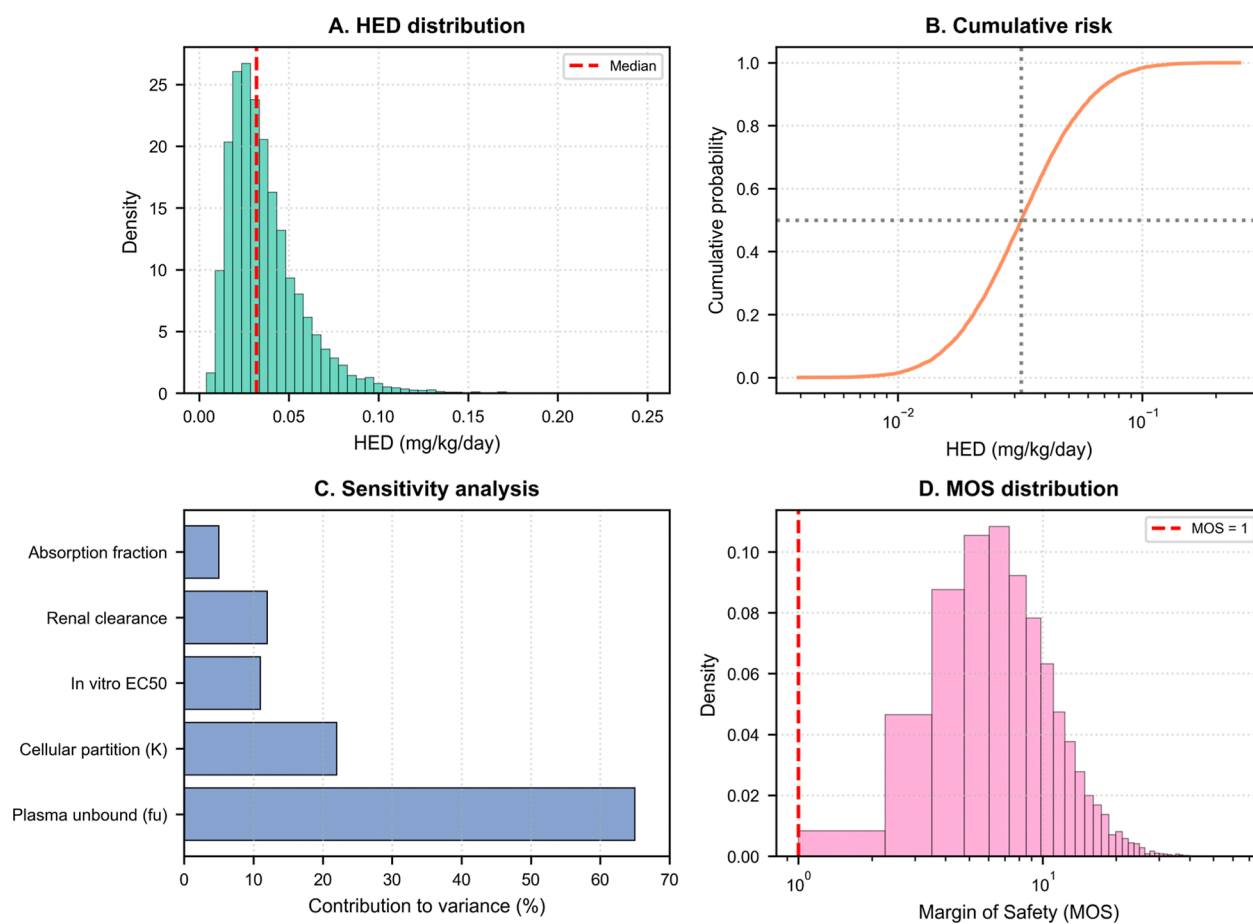


Figure 4. Integrated framework predictions and probabilistic risk assessment of neodymium nitrate. (A) Probability distribution of human equivalent dose (HED) predicted by the integrated AIVIVE-PBPK-QIVIVE framework (10,000 Monte Carlo iterations). The distribution is right-skewed (skewness = 1.85) with a median of 0.032 mg/kg/day (95% CI: 0.012–0.098 mg/kg/day) and mode of 0.026 mg/kg/day. The red dashed line marks the median; the shaded area represents the 95% confidence interval. Vertical dotted lines indicate risk thresholds: <0.01 mg/kg/day (very low risk), 0.1–0.2 mg/kg/day (high risk), and ≥ 0.2 mg/kg/day (very high risk). (B) Cumulative probability distribution of HED. The plot shows the probability that HED exceeds a given value. Key percentiles: $P(\text{HED} < 0.01 \text{ mg/kg/day}) = 12\%$, $P(\text{HED} < 0.024 \text{ mg/kg/day}) = 45\%$, $P(\text{HED} < 0.037 \text{ mg/kg/day}) = 63\%$, $P(\text{HED} > 0.1 \text{ mg/kg/day}) = 18\%$, and $P(\text{HED} > 0.2 \text{ mg/kg/day}) = 7\%$. (C) Global sensitivity analysis (Sobol indices) quantifying the contribution of key input parameters to HED variance. Plasma protein binding (f_u) is the primary source (65%), followed by cellular partition coefficient (K , 22%), renal clearance (CL, 12%), absorption fraction (F_{abs} , 5%), and in vitro EC_{50} (11%). (D) Margin of safety (MOS) distribution relative to a reference dose ($\text{RfD} = 0.25 \text{ mg/kg/day}$). The median MOS is 7.8; $P(\text{MOS} < 1) = 4.2\%$ (unsafe exposure). 58 percent of simulations have $\text{MOS} < 10$ (insufficient safety margin), while 12% have $\text{MOS} > 100$ (very safe). The red vertical line marks $\text{MOS} = 1$.

Table 2. Compiled Neodymium Exposure Estimates for Different Population Groups

population group	exposure pathway	estimated daily intake ($\mu\text{g/kg/day}$)	basis/assumptions
general population (China)	dietary	0.65 (central estimate)	$250.9 \mu\text{g total REEs/day} \times 18\% \text{ Nd} \div 70 \text{ kg}$
residents in mining area (e.g., Baotou)	multimedia (Total)	16.7 (central estimate)	$6.5 \text{ mg total REEs/day} \times 18\% \text{ Nd} \div 70 \text{ kg}$

$\text{MOE} < 100$, strongly indicating an insufficient safety margin for this subpopulation, highlighting a potential priority for monitoring and risk management. Sensitivity analysis confirmed that HED uncertainty (dominated by f_u and K) contributed $>90\%$ to MOE variance, underscoring the need for refined TK data over exposure assessment to reduce overall uncertainty.

To further contextualize the risk assessment, predicted HED values under three hypothetical exposure scenarios—occupational, environmental via drinking water, and accidental acute

release—were compared with the reference dose ($\text{RfD} = 0.25 \text{ mg/kg/day}$). As illustrated in Figure 6, the predicted HED for the accidental release scenario exceeds the high-risk threshold by a substantial margin, whereas occupational and environmental drinking water exposures remain well below the RfD . Error bars represent 95% confidence intervals derived from Monte Carlo simulations.

3.6. Implications and Regulatory Application

The principal innovation lies in the closed-loop, model-informed workflow itself, where HTTK serves a diagnostic role in prioritizing targeted experiments, the results of which directly calibrate and reduce uncertainty in the final risk estimates. This study presents significant methodological advances: the first AIVIVE framework for REEs, a systematic approach to adapt HTTK for metals via calibration, and the integration of AIVIVE, PBPK, and QIVIVE into an end-to-end, mechanism-based workflow that aligns with the 3Rs principle.

The nonlinear kinetics of neodymium nitrate have critical implications; traditional constant-parameter assumptions would lead to inaccurate risk estimates. Our framework incorporates

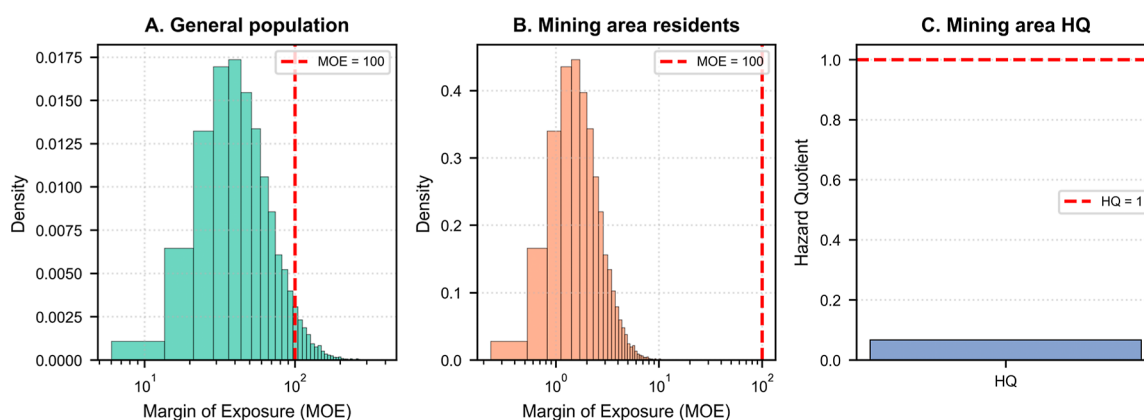


Figure 5. Probabilistic risk characterization for different exposure scenarios. (A) Cumulative probability distributions of the margin of exposure (MOE) for the general Chinese population (dietary intake, blue curve) and residents in rare earth mining areas (high exposure, orange curve). $MOE = HED/exposure\ dose$. The vertical dashed line indicates $MOE = 100$, a common benchmark for negligible concern. For the general population, the median MOE is 49.2 and $P(MOE < 100) = 96.8\%$. For mining area residents, the median MOE is 1.9 and $P(MOE < 100) = 100\%$, indicating an insufficient safety margin. (B) Probability distribution of the Hazard Quotient (HQ) for mining area residents ($HQ = exposure\ dose/RfD$). All simulated HQ values exceed 0.05, and 100% exceed 0.1, confirming elevated risk. (C) Comparison of MOE central estimates (medians) and uncertainty ranges (95% CI shown as boxplots) for the two population groups. Note the logarithmic scale on the y-axis. The mining group exhibits both a much lower MOE and a wider uncertainty interval, underscoring the need for refined exposure and toxicokinetic data in this subpopulation.

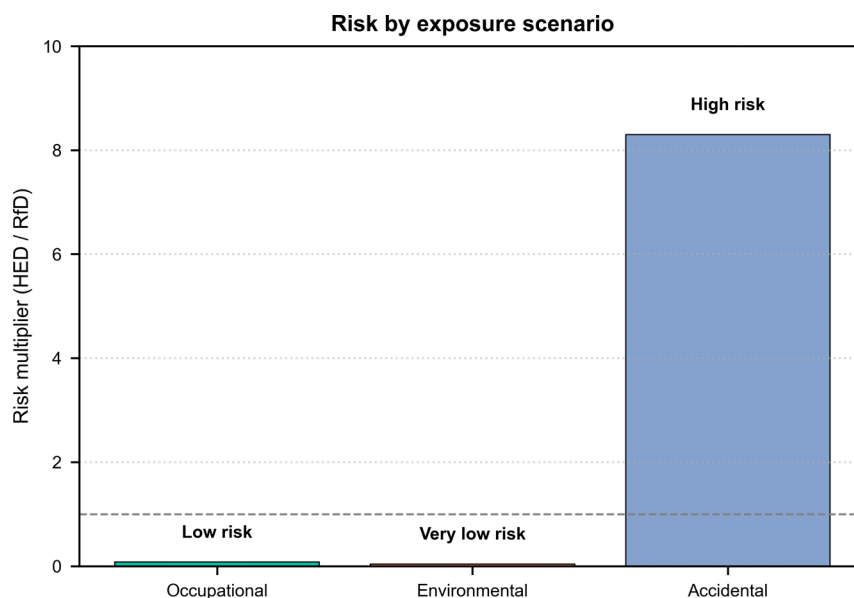


Figure 6. Risk characterization of neodymium nitrate under different exposure scenarios. Bar chart showing the predicted human equivalent dose (HED) for three exposure scenarios: occupational (air concentration 0.1 mg/m³, daily intake 0.002 mg/kg/day), environmental via drinking water (0.05 mg/L, 0.001 mg/kg/day), and accidental acute release (10 mg/m³, 0.2 mg/kg/day). The red line with circular markers indicates the risk multiplier (HED divided by the reference dose $RfD = 0.25\ mg/kg/day$). Risk categories are defined as “low risk” (multiplier < 0.1), “very low risk” (multiplier < 0.05), and “high risk” (multiplier > 8). Error bars represent 95% confidence intervals of the HED estimates derived from Monte Carlo simulations.

this complexity for more physiologically realistic modeling. The challenges identified with HHTK for neodymium nitrate likely extend to other metal compounds, and our calibration/hybrid parameter strategy provides an adaptable template.

The probabilistic framework offers clear advantages over deterministic point estimates by explicitly quantifying uncertainty, enabling more informed decision-making. Based on the results, we recommend an 8 h time-weighted average occupational exposure limit (OEL-TWA) of 0.05 mg/m³ for neodymium nitrate.

3.7. Limitations and Future Directions

Limitations include the current focus on oral exposure to soluble Nd salts; extension to other routes (inhalation and dermal) and REE compounds is needed. The characterization of nonlinear TK parameters (f_u and K) is based on a limited number of concentration points. While the data strongly suggest concentration-dependence, future studies with finer concentration resolution are needed to fully validate the shape and mechanism of these relationships. Additional metal-specific processes (active transport and subcellular accumulation) could be integrated. Validation with human data would strengthen confidence. There is a paucity of systematic biomonitoring

data for REEs to refine exposure estimates. While the PBPK model was parametrized for human physiology, the final HED calculation employed body surface area (BSA) scaling from mouse-derived clearance parameters, a standard regulatory default. Future integration of in vitro to in vivo extrapolated (IVIVE) human clearance parameters into the PBPK model could refine interspecies extrapolation and reduce the reliance on allometric scaling. The PBPK model was validated against tissue distribution data from a high-dose study, which was necessary for analytical quantification. While the model's physiological structure provides confidence in its extrapolative power, the accuracy of predictions at environmental exposure levels (orders of magnitude lower) remains to be experimentally confirmed. This extrapolation uncertainty is partially captured in the probabilistic Monte Carlo simulations. Future work should extend the framework to other metals/REEs, incorporate additional omics layers, develop standardized calibration protocols, and create user-friendly software tools.

4. CONCLUSION

We successfully developed and validated an integrated AIVIVE-PBPK-QIVIVE framework with HTKK validation for the probabilistic risk assessment of neodymium nitrate. Key findings include: (1) the AIVIVE model accurately predicted toxicity pathways; (2) experimental characterization revealed significant nonlinear kinetic behaviors; (3) standard HTKK models required compound-specific calibration for metals; (4) the integrated framework predicted a median HED of 0.032 mg/kg/day, with an 18% probability of high-risk exposure; and (5) sensitivity analysis identified plasma protein binding and cellular partition coefficients as primary uncertainty sources.

This work establishes a novel paradigm for the mechanism-based, quantitatively robust risk assessment of data-limited compounds, particularly relevant for metals with complex kinetics. It supports the transition toward animal-free testing and provides a scientific basis for regulatory decision-making. By integrating probabilistic hazards with real-world exposure, the framework enables context-aware risk quantification, identifying the general population as low risk but flagging specific high-exposure subpopulations for targeted management.

■ ASSOCIATED CONTENT

Data Availability Statement

All data generated or analyzed are included in this article and its Supporting Information. The transcriptomic data set is available in the National Genomics Data Center (NGDC) under accession no. OMIX010050. Model code is available at <https://github.com/SHCDC-LAB>.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.chemrestox.6c00103>.

Detailed experimental protocols; AIVIVE model architecture and training details; mathematical derivation of nonlinear binding model; PBPK model equations, parameters, and structure; QIVIVE and reverse dosimetry algorithm; probabilistic assessment methodology and parameter distributions; DEG lists, pathway results, raw experimental data, statistical summaries, sensitivity indices, and BMD values; and validation plots for PBPK model (PDF)

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editing. Da-sheng Lu: writing—review and editing. All authors have read and agreed to the published version of the manuscript.

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Notes

Animal experiments were approved by the Experimental Animal Ethics Committee of Shanghai Municipal Center for Disease Control and Prevention (IACUC-PZ-035–2025). All procedures followed relevant guidelines and regulations.

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